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HydroCraft: an efficient online platform for hydrological modeling integrated with a large language model agent

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ABSTRACT

Hydrological modeling is essential for water resources management but often requires specialized expertise, creating a significant barrier to its broad application. To address this challenge, we developed HydroCraft, a web-based platform designed to democratize the modeling process. HydroCraft provides a comprehensive and integrated workflow for basin extraction, driving data generation, parameter calibration, and result visualization, enabling rapid modeling for any watershed globally. Key features of the platform include: an adaptive watershed delineation method that optimizes the distribution of computational units based on user-defined sub-basin area thresholds; the integration of three hydrological models (THREW, Xin'anjiang, and Miyun Hydrological Model) suitable for diverse climatic and geological conditions; and the incorporation of a Large Language Model (LLM) agent, which serves as a robust assistant to provide guidance and automated task execution. Users only need a device with a web browser and an internet connection to perform advanced hydrologic simulations. Two case studies (user-driven vs. LLM-agent-assisted) demonstrated the platform's efficiency and intelligence. The platform advances hydrological modeling toward greater intelligence and accessibility, providing support for applications such as water resource management and flood forecasting.

1. Introduction

Hydrological models serve as essential tools for analyzing the water cycle, playing a critical role in flood forecasting, water resource management, and related fields. The International Association of Hydrological Sciences (IAHS) "HELPING" scientific decade (2023–2032) underscores the urgent need to bridge global water science with local actionable solutions, emphasizing co-creation and innovative methodologies to achieve sustainable water management. In this context, advancing hydrological modeling tools towards greater intelligence, accessibility, and integration is paramount for translating scientific insight into effective design and decision-making.

The progression of hydrological modeling tools is deeply intertwined with advancements in computing technology. This trajectory, traceable to the closed standalone systems of the 1980s, through modular open-source ecosystems, to the current cloud-based collaborative platforms, has fundamentally transformed methodologies in water science research. Early tools, represented by HEC-HMS and MIKE SHE (Abbott et al., 1986), utilized native compilation techniques. They standardized the mathematical representation of hydrological processes and introduced graphical user interfaces to replace command-line operations. Subsequent iterations significantly lowered the barrier to numerical simulation. The widespread adoption of scripting languages like Python and MATLAB in the 21st century ushered in a modular, open-source era for hydrological modeling. By the mid-2000s, the integration of the Soil and Water Assessment Tool (SWAT) model with Geographic Information Systems (GIS) gradually established it as one of the popular and standard tools for watershed hydrological and water quality simulation globally (Arnold et al., 1998; Arnold et al., 2012; Bieger et al., 2017). Tools like FloPy (Bakker et al., 2016) further advanced this trend by enabling users to control solver kernels (e.g., MODFLOW) via scripts, promoting a shift from "black-box" operations towards customizable modeling workflows.

Despite these advancements, constructing hydrological models remains a cumbersome, time-consuming process with a high dependency on specialized expertise, which impedes their broader application. Hydrological modeling is essentially a complex system engineering task that translates physical processes into mathematical formulations, encompassing data preprocessing, computational unit discretization, input data preparation, parameter calibration, and validation. Although

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computing advancements have spurred the creation of various modeling aids, significant practical shortcomings persist. First, fragmented toolchains reduce modeling efficiency. Predominant desktop software applications often operate in isolation, forcing researchers to switch repeatedly between different tools to complete the entire workflow—from geospatial data processing to model parameter optimization. This disjointed approach is operationally complex, inefficient, and increases the risk of error propagation. Second, the high expertise threshold limits model adoption. Complex parameters and terminology present a significant barrier for non-expert users. In underserved regions lacking hydrological specialists, users may misinterpret technical terms or model principles, leading to incorrect parameter settings or, ultimately, abandonment of hydrological models in favor of less rigorous empirical methods for water-related hazard assessment. Finally, the local deployment paradigm of traditional software constrains scalability and collaboration. Dependencies on specific operating systems, large component libraries, and limited local computational resources not only complicate setup and increase instability but also confine modeling tasks to individual machines, preventing leverage of cloud resources and computational power for enhanced productivity.

Web-based platforms have emerged to lower these barriers. The proliferation of remote sensing and reanalysis techniques has led to the open availability of extensive datasets covering topography, meteorology, land cover, and hydrology. This data openness enhances research accessibility and facilitates application development. Numerous web-based hydrological models and modeling tools have subsequently emerged, aiming to provide user-friendly solutions, particularly for those with limited modeling capacity. Recent tools like HydroShare (Tarboton et al., 2014; https://www.hydroshare.org), WikiWatersheds (https://wikiwatershed.org), StreamStats (Ries et al., 2017; https://streamstats.usgs.gov/ss/), WEPPcloud (Lew et al., 2022; https://wepp.cloud/weppcloud), LakeShed (Zhang et al., 2025; https://lakewatershed.com/main) and PLUVIO-TECHNOPOLE (Grimaldi et al., 2025; https://www.hydrolab.unitus.it/technopole/#/dashboard) assist users with tasks such as watershed delineation and data retrieval from public databases. These tools, however, are not without their limitations. They tend to offer only niche functionality, supporting isolated steps within the modeling process or focusing solely on data sharing/acquisition rather than providing an integrated workflow. Furthermore, their applicability is often confined to specific geographical areas (e.g., the continental United States, Italy), limiting their global applicability.

Recent advances in large language models (LLMs) have stimulated growing interest in their application to hydrology and water resources management. Existing studies have explored LLMs for hydrological knowledge representation, scientific question answering, workflow automation, decision support, and model calibration. For example, WaterGPT (Ren et al., 2024) developed a hydrology-oriented LLM and multi-agent framework capable of handling multimodal hydrological tasks such as waterbody extraction, hydrological question answering, and intelligent decision-making, demonstrating the potential of domain-specific LLMs in water resources applications. Similarly, HydroLLM (Kizilkaya et al., 2025) explored the construction of a domain-specific hydrological language model through fine-tuning on hydrology-focused question–answer datasets, highlighting the importance of domain adaptation and specialized corpora for improving hydrological reasoning and contextual understanding.

Several recent studies have also focused on integrating LLMs with hydrological modeling workflows and scientific computing environments. HydroSuite-AI (Pursnani et al., 2024) introduced an LLM-driven framework combining hydrological libraries, automated code generation, workflow assistance, and documentation support to facilitate hydrological research and model development. IWMS-LLM (He et al., 2025) further explored intelligent water resources management by integrating language models with hydrological information systems and decision-support workflows. In addition, AQUAH (Yan et al., 2025) proposed an end-to-end language-based hydrological agent capable of automatically retrieving datasets, configuring hydrological models, running simulations, and generating analyst-ready reports directly from natural-language prompts.

Beyond workflow automation, recent work has increasingly explored the use of LLMs as autonomous agents for hydrological simulation and calibration. Hydro-Agent introduced a physics-integrated multi-agent framework that autonomously generates, debugs, and executes groundwater inverse-modeling workflows while maintaining physical consistency and interpretability (Ma et al., 2026). Similarly, Zhu et al. (2026) investigated the feasibility of using LLMs as calibration agents for hydrological modeling and demonstrated that advanced LLMs can achieve efficient parameter optimization with physically interpretable parameter sets.

These studies indicate that LLMs are rapidly evolving from general-purpose conversational tools toward domain-specific intelligent assistants and autonomous agents in hydrology. Current applications span hydrological knowledge modeling, multimodal data analysis, automated coding, hydrological simulation, workflow orchestration, and parameter calibration. However, most existing studies focus on isolated tasks such as question answering, workflow automation, or autonomous model execution, while relatively limited attention has been paid to integrated and user-friendly hydrological modeling platforms that combine conversational interaction, automated workflow management, hydrological model execution, and reproducible scientific analysis within a unified framework.

Against this backdrop, this study develops HydroCraft (www.hydrocraft.hydr.tsinghua.edu.cn), an online hydrological modeling platform integrated with an LLM agent. HydroCraft is designed to address the limitations identified above by providing a complete and globally applicable modeling workflow, from adaptive watershed delineation and multi-source data preparation to parameter optimization, visualization, and model sharing. Crucially, HydroCraft supports a flexible, human-in-the-loop modeling paradigm: users can execute the workflow manually through the graphical web interface or delegate selected tasks to the LLM agent through natural-language instructions. This approach establishes an adaptive intelligent workbench that aims to democratize advanced hydrological modeling while preserving user agency and supporting the collaborative, solution-oriented vision of modern hydrology.

2. Methodology

2.1. Watershed delineation

In distributed hydrological modeling, determining the scale at which heterogeneity is averaged and the optimal heterogeneity level within modeling units is crucial for filtering irrelevant details (Sivapalan, 2003; Didszun & Uhlenbrook, 2008). The Representative Elementary Area (REA) concept (Wood et al., 1988) provides a theoretical framework for defining a threshold area where hydrological responses become homogeneous, reducing the impacts of spatial variability on runoff. However, REA values are process-dependent and scale-sensitive, precluding a universal threshold (Seyfried & Wilcox, 1995). Subsequent research demonstrates that distributed model performance depends on interactions between modeling scale, input data resolution/quality, model architecture, and watershed characteristics (e.g., Molnar & Julien, 2000; Vazquez et al., 2002; Rojas et al., 2008; Shrestha et al., 2006; Merz et al., 2009). Additionally, spatiotemporal scale alignment is crucial as mismatched scales can invalidate empirical formulations (Melsen et al., 2016). An appropriate modeling scale is therefore essential: oversimplified scales risk model distortion, while overly complex scales waste computational resources, with potential issues from data noise and parameter scale-dependency also influencing performance.

To address these scale challenges, this study proposes an adaptive watershed extraction and subdivision strategy. Users define upper and

lower thresholds for sub-watershed areas. The workflow (as shown in Fig. 1) starts conventionally: DEM depression filling, D8 flow direction calculation, and flow accumulation computation. A user-defined flow accumulation threshold identifies stream cells and thus controls river network density. Applying the "stream link" tool assigns unique IDs to stream segments between confluences. The "watershed" tool then uses these IDs to delineate sub-watersheds corresponding to each segment. The number and shape of these initial sub-watersheds depend solely on confluences defined by the flow accumulation threshold and terrain complexity, often resulting in sub-watersheds that are excessively large in uniform areas and excessively small in complex terrain.

An adaptive subdivision phase corrects these extremes using user-defined maximum and minimum allowable sub-watershed areas:

(1) Subdividing Oversized Units: Sub-watersheds exceeding the maximum area are identified. Their corresponding stream link ID is split to create two new segments, triggering watershed re-delineation to produce two smaller sub-watersheds. This process repeats iteratively until all sub-watersheds are below the maximum area threshold.
(2) Merging Undersized Units: Sub-watersheds below the minimum area (typically arising between close confluences) are identified and merged with their immediate downstream neighbor. Combined areas are recalculated. This process repeats iteratively until all sub-watersheds meet or exceed the minimum area.

The strategy thus achieves a more uniform sub-watershed area distribution, enhancing computational efficiency and accuracy for hydrological modeling while preserving hydrological process continuity.

Fig. 2 compares the results of the adaptive watershed delineation method with of the conventional approach. Under the conventional method (Fig. 2, left), the watershed is divided into 13 sub-watersheds. Among them, Sub-watershed 6 has a noticeably smaller area than the others, while Sub-watershed 1 is significantly larger in size.

In the adaptive delineation method, Sub-watershed 1 is first selected for further subdivision. The stream link raster segment representing Sub-watershed 1 is split in half: the downstream portion retains the ID 1, while the upstream portion is assigned a new ID—14 (equal to the current maximum ID plus one). A new watershed delineation is then performed. Since the newly created Sub-watershed 14 still exceeds the maximum area threshold, it is further subdivided using the same approach, resulting in Sub-watersheds 14 and 15.

For the undersized Sub-watershed 6, it is merged with its downstream neighbor, Sub-watershed 8, to form a single sub-watershed. It is worth noting that Sub-watershed 8 (and its corresponding stream segment) has three upstream tributaries (Sub-watersheds 2, 3, and 5). When multiple sub-watersheds are merged, the number of upstream tributaries for the resulting sub-watershed may increase. In contrast, Sub-watersheds 1 and 14 each have only one upstream tributary. Appendix A provides a detailed evaluation and sensitivity analysis of the adaptive watershed extraction method in real-world watersheds. The results indicate that, in terms of the sub-basin area distribution, the adaptive method produces a more concentrated distribution with lower variability, whereas the conventional method yields a broader and more dispersed distribution, generating sub-basins of anomalous sizes.

2.2. Hydrological models

Three hydrological models, namely the THREW (Tsinghua Hydrological Model based on Representative Elementary Watershed, Tian

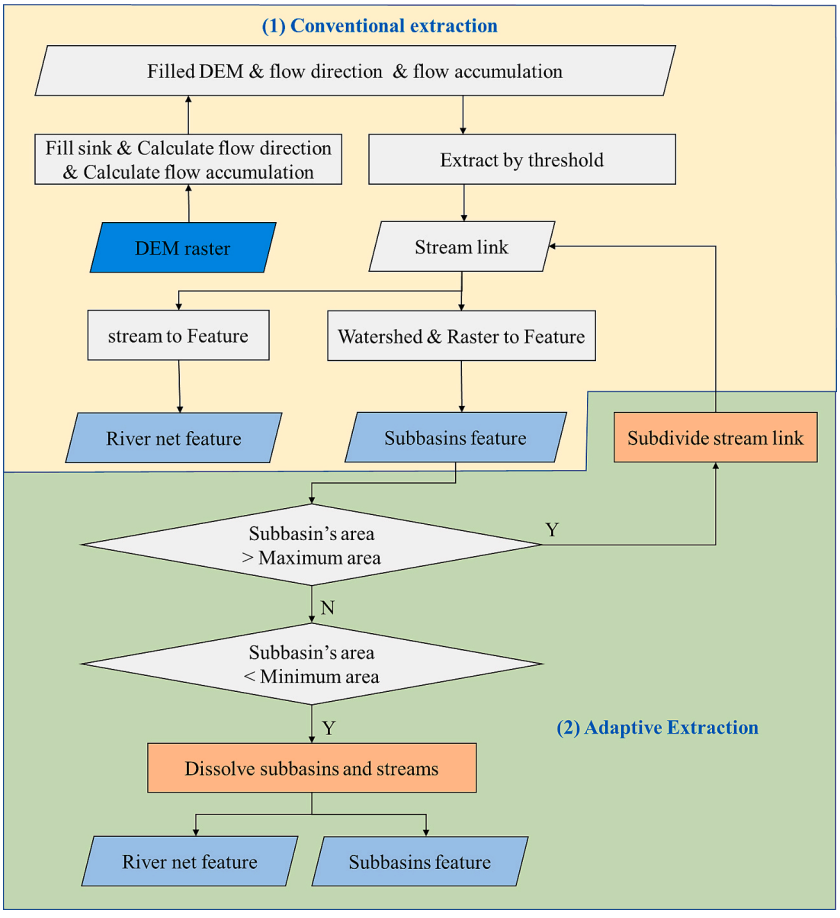


Fig. 1. Flowchart of the adaptive watershed extraction method based on user-defined sub-watershed area thresholds.

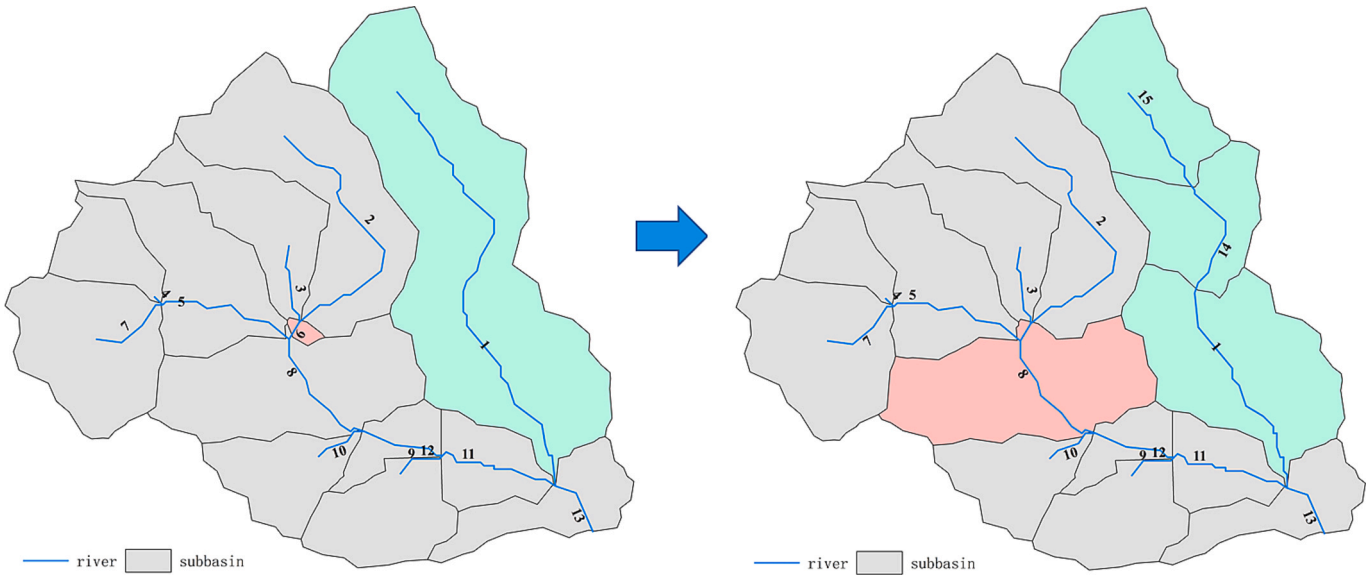


Fig. 2. Comparison of adaptive vs conventional watershed extraction results.

et al., 2006), Xin'anjiang (Ren-Jun 1992), and Miyun Hydrological model (Tudaji et al., 2025a), have been integrated into this platform. THREW integrates a set of equilibrium equations for mass, momentum, energy, and entropy, along with constitutive relationships governing various fluxes between representative units and sub-regions within units. The THREW model vertically partitions the Representative Elementary Watershed (REW) into two distinct strata: the surface and subsurface layers. Specifically, the surface layer is discretized into six zones: bare soil (b-zone), vegetated (v-zone), snow covered (n-zone), glacier covered (g-zone), sub-stream-network (t-zone), and main channel reach (r-zone). Beneath this, the subsurface layer is divided into the unsaturated (u-zone) and saturated (s-zone) zones. This zoning strategy allows for the explicit representation of key landscape features within the REW framework (Tian et al., 2008). The model's robustness has been validated across diverse climatic and geological settings, with successful applications in the Han River basin (Sun et al., 2014; Li et al., 2018), the Yarlung Tsangpo-Brahmaputra River basin (Xu et al., 2019; Nan et al., 2021; Cui et al., 2023), and various medium-to-small basins in Southeast China (Tudaji et al., 2025b).The structure of runoff generation processes within the THREW model and the parameters requiring calibration are depicted in Fig. 3 and Table 1, respectively.

The Xin'anjiang model is a classic hydrological model that has been widely applied, especially in the southern China (e.g., Jayawardena and Zhou, 2000; Li et al., 2009; Jiang et al., 2023). The Miyun Hydrological Model was further developed from the Xin'anjiang model for the semi-humid soil-rock mountainous areas of northern China. It has demonstrated good performance on both daily and hourly timescales in multiple northern basins (Tudaji et al., 2025a). Detailed descriptions of

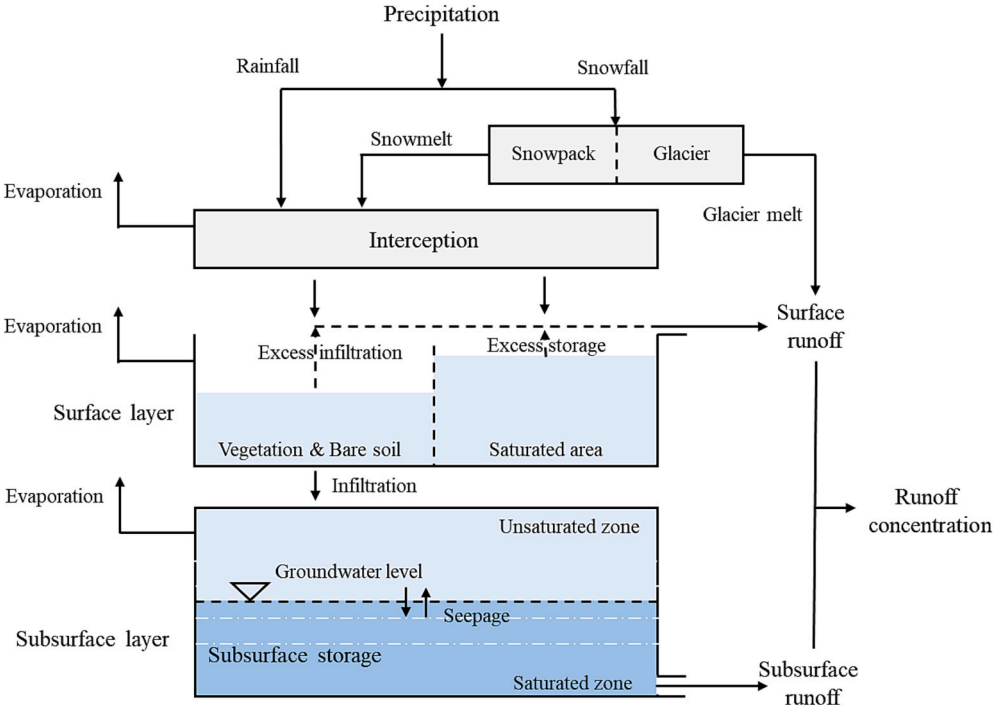


Fig. 3. Structural diagram of runoff generation processes in the THREW model.

Table 1
Parameters to be calibrated in the THREW model.

Symbol	Unit	Physical descriptions	Range
Kv	–	fraction of potential transpiration rate over potential evaporation	0–0.8
nt	–	Manning roughness coefficient for hillslope	0–0.2
GaIFL	–	coefficient for spatial averaged infiltration capacity	0–0.7
GaEFL	–	coefficient for spatial averaged exfiltration capacity	0–0.7
WM	cm	Tension water storage capacity, which was used in the Xinanjiang model (Zhao et al., 1992) to calculate saturation area	0–10
B	–	Shape coefficient used in Xinanjiang model to calculate saturation area	0–1
KKA	–	Coefficient to calculate subsurface runoff in $R_g = KKD \cdot S \cdot K_S^S \cdot (y_s/Z)^{KKA}$, where S is the topographic slope, K_S^S is the saturated hydraulic conductivity, y_s is the depth of saturated ground- water and Z is the total soil depth	0–6
KKD	–	See description for KKA	0–0.5
C_1	–	Coefficient to calculate the runoff concentration process using the Muskingum method: $Q_2 = C_1 \cdot I_1 + C_2 \cdot I_2 + C_3 \cdot Q_1 + C_4 \cdot Q_{lat}$, where I_1 and Q_1 are the inflow and outflow at the prior step, respectively; I_2 and Q_2 are the inflow and outflow at the current step, respectively; Q_{lat} is lateral flow of the river channel; $C_3 = 1\text{-}C_1\text{-}C_2$; and $C_4 = C_1 + C_2$	0–1
C_2	–	See description for C_1	0–1

these two models are available in the HydroCraft platform documentation.

2.3. LLM agent

LLM agents are autonomous computational entities that enhance the performance of their underlying LLMs by integrating memory, knowledge augmentation, and planning to tackle diverse tasks (Sun et al, 2025; Huang et al., 2024). The memory system utilizes long-term memory for foundational knowledge, short-term memory for real-time contextual awareness, and active memory management to maintain information relevance. For knowledge augmentation, agents leverage techniques like Retrieval-Augmented Generation (RAG) to integrate external knowledge bases or invoke tools such as APIs and databases, thereby overcoming the inherent knowledge limitations and timeliness issues of LLMs. Furthermore, their planning capabilities enable them to decompose complex goals, simulate state transitions, and optimize action sequences. The synergy of these components empowers agents to autonomously manage dynamic, multi-step tasks beyond simple instruction execution, demonstrating significant potential for autonomous task planning and execution. This makes them highly promising for applications in scientific research, automation, education, and beyond.

In this study, we developed a LLM agent based on ChatGLM (Generative Language Model, GLM Team, 2024), designed to provide comprehensive intelligent services through precise question-answering and autonomous task execution. ChatGLM is an intelligent dialogue system based on the powerful language model GLM-4, with natural language understanding and generation capabilities. The high performance of this agent is primarily attributed to its integration of RAG technology. RAG serves as a key technique to address the inherent limitations of LLMs, significantly enhancing the accuracy and reliability of model outputs through a two-stage paradigm of retrieval and generation.

During the retrieval phase, RAG leverages a vector database to efficiently match the most relevant document snippets from a private knowledge base to the user's query. In the generation phase, these retrieved contextual snippets are fed into the LLM to guide the generation of fact-based, precise responses. The core advantages of RAG technology are threefold: first, it effectively mitigates the model's "hallucination" problem, ensuring consistency between answers and the knowledge base; second, it allows the knowledge base to be updated manually by developers (e.g., with new or corrected documents) to guarantee information timeliness, without requiring retraining or fine-tuning of the underlying LLM model; third, by enabling the private deployment of the knowledge base, it reduces reliance on general-purpose models, offering significant benefits in terms of privacy protection and cost control.

The information processing workflow of the ChatGLM agent, powered by RAG, is detailed as follows (also shown in Fig. 4):

1. Knowledge base construction and indexing

Data ingestion: raw materials are imported into the system via file uploads or provided links, serving as the data source for the knowledge base.

Text segmentation and vectorization: the text within the knowledge base is segmented into semantic units and subsequently converted into high-dimensional vectors using an embedding model to facilitate efficient semantic retrieval.

Index construction: the generated vectors are stored in a specialized vector database, where an index is built to enable rapid Approximate Nearest Neighbor (ANN) search.

2. Retrieval phase

Query preprocessing: upon receiving a user query, the system converts it into its corresponding vector representation.

Relevance matching: the system performs an ANN search within the vector index to retrieve the text snippets that are semantically most similar to the query vector.

3. Generation phase

Context integration: the retrieved relevant text snippets are incorporated as augmented context, which is then combined with the system prompt and dialogue history to form the final input prompt.

Response generation: the LLM utilizes this integrated prompt to generate the final response, ensuring it is both aligned with the user's intent and grounded in reliable knowledge sources.

3. Implementation of the platform

3.1. Platform architecture

To ensure accessibility and cross-platform compatibility, the platform is developed using a web-based client–server (C/S) architecture. Users can directly access the front-end interface via a web browser, while the back-end services are built on the.NET framework to support high-concurrency task processing and distributed resource scheduling.

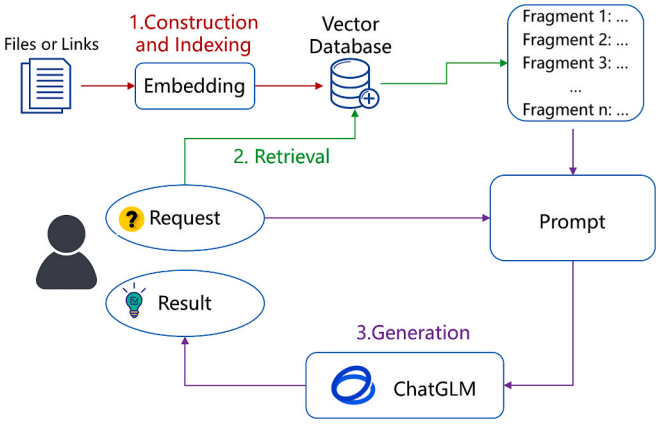


Fig. 4. Information processing workflow for the RAG-based ChatGLM Agent.

The overall architecture is guided by the core design principles of modularity and extensibility, integrating the entire workflow of hydrological modeling with intelligent services. The platform architecture consists of five layers (as shown in Fig. 5).

(1) View Layer: This front-end layer consists of user interface (UI) elements such as the map layout, toolbar, dialog boxes, and progress bars. Its primary functions include visualizing spatial data, listing available tools, facilitating user-tool interactions, and displaying task progress and results.
(2) Logic Layer: Acting as a bridge between the View and Service layers, this layer manages user interactions. It receives user requests from the front-end, forwards them to the back-end server, and relays the processing results back to the View Layer.
(3) Service Layer: This layer is responsible for executing the core business logic, with specific tasks including watershed extraction, data format conversion, model-driven data generation, model parameter optimization, and providing AI agent services.
(4) Data Layer: Leveraging a geographic data processing engine and a document management system, this layer is responsible for storing and managing multi-source heterogeneous data. The main data types include: (1) Base modeling data, such as topography, meteorology, vegetation, and soil data; (2) Model and Algorithm Library, containing pre-configured model templates and various executable programs or scripts; (3) Frontend Resource Library, such as spatial data of river channels and hydrological stations, along with icon resources; (4) Result files generated after the execution of user tasks.
(5) Base Layer: Serving as the physical foundation of the entire architecture, this layer comprises hardware infrastructure, including storage devices, computing servers, and network systems, which provide the underlying support for the platform's stable operation.

3.2. Implementation of the modules

3.2.1. Watershed delineation

The specific method for watershed delineation has been described in detail in Section 2.1. The back-end service for this function primarily utilizes the spatial analysis module of arcpy. The back-end server invokes the Python interpreter to execute the Python script for watershed extraction. Required user inputs include the watershed outlet, selected DEM data source, watershed boundary, outlet snapping tolerance, flow accumulation threshold, and maximum/minimum sub-watershed areas. Table 2 lists the DEM datasets integrated into the platform for watershed extraction, encompassing several global-scale DEMs at 90 m and 30 m resolutions. This collection enables the construction of hydrological models for any watershed globally.

To enhance user experience, a simplified watershed extraction method was designed, requiring only the watershed outlet coordinates. However, performing operations over the full DEM extent without a specified area is inefficient. To address this, two optimization strategies are employed. First, the system employs the HydroBASINS dataset (Lehner and Grill, 2013) to identify a preliminary upstream boundary for the target outlet, which then delimits the operational area for watershed extraction. Second, pre-computed flow direction and flow accumulation rasters from the HydroSheds dataset (Lehner et al., 2008) are used to accelerate processing. In this mode, the default outlet snapping tolerance is set to 0.01°, the flow accumulation threshold is 1/20th of the total area of the watershed of interest, and both the maximum and minimum sub-watershed areas are set to zero (i.e., only basic watershed extraction and delineation are performed without further subdivision and dissolving).

3.2.2. Driving data generation

This module is designed to calculate the average meteorological and basin attribute data for each computational unit (i.e., sub-basin) to generate the driving data for the hydrological model. Its implementation process is similar to that of the basin extraction function: the front-end view layer collects user parameters and transmits them to the server, which then invokes Python scripts to calculate the average data for each

Table 2
List of DEM datasets integrated in the platform.

Order	Dataset name	Resolution (m)	Coverage area	Source
1	HydroSheds	90	Globe	Lehner et al., 2008
2	MERIT_DEM	90	Globe	Yamazaki et al., 2017
3	SRTM_DEM_V4	90	Globe	Jarvis et al., 2008
4	ALOS	30	Globe	T. Tadono et al., 2014

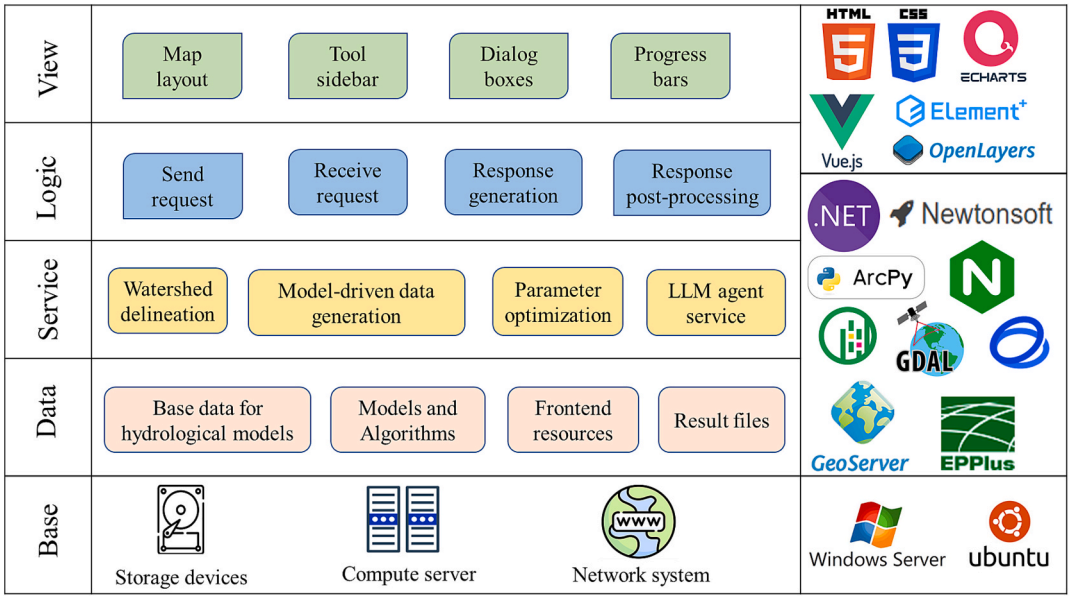


Fig. 5. Architecture of the HydroCraft platform and key technologies.

sub-basin based on the database. To facilitate direct editing and programmatic read/write operations, all model input data are stored in text format. The resulting driving data file is an n-by-m matrix, where the number of rows (n) corresponds to the length of the time series (arranged chronologically), and the number of columns (m) corresponds to the number of sub-basins (arranged in ascending order of Subbasin_ID).

This module offers two implementation methods. The first is the "Zonal Data Statistics" tool in the toolbar, which calculates the average time series for a single data type (e.g., precipitation, evaporation). The output file format is fully compatible with the model's input requirements and can be directly used as driving data. The second method is the "Prepare All Driving Data" tool, which generates all required data for the model in a single operation (including precipitation, evaporation, temperature, NDVI, LAI, and soil data) and packages them into a complete hydrological model archive. This archive contains the model executable, default initial state and parameter files, and all the aforementioned driving data files. The model can be executed directly in any environment with the.NET runtime installed.

The list of currently integrated driving datasets is provided in Table 3 (subject to continuous updates). To support hydrological simulations in any catchment globally, at least one globally-covered dataset is included for each data type. Given the critical role of precipitation in simulations, a variety of precipitation datasets have been specifically integrated for user selection.

3.2.3. Model parameter calibration

Parameter calibration is a critical step in hydrological modeling, aimed at adjusting model parameters to accurately reflect the specific characteristics of a catchment, including its topography, soil, vegetation, climate, and hydrogeology. The initial model generated through "Basin Extraction" and "Driving Data Preparation" is initialized with default template parameters, which may not be suitable for all target catchments. Therefore, the calibration process enhances the model's predictive accuracy and reliability by optimizing these parameters.

This platform integrates a "Model Parameter Calibration" module, employing the NSGA-II algorithm (Deb et al., 2002) for parameter optimization. Renowned for its efficiency, robustness, and excellent distribution of the solution set, NSGA-II is widely applied in fields such as hydrological model calibration and engineering design optimization, providing a powerful solution for complex multi-objective optimization problems.

To facilitate execution, the NSGA-II optimization algorithm is compiled into a standalone executable file. Prior to performing hydrological model parameter optimization, a configuration file must be set up with the following parameters: the length of the observed streamflow data series, the warm-up period length, the maximum number of iterations (default: 2000), the desired maximum value for the optimization objective (NSE, default: 0.95), the number of model parameters, and the relative paths to relevant files (model parameter file, model executable, model output file, and observed streamflow file). For the specific models integrated on this platform, some key settings, such as the number of parameters, their value ranges, and file paths, are preset. The length of the observed streamflow data series is determined by the user-defined simulation time range and calculation step, and is automatically configured by the backend system based on the submitted data.

From the user's perspective, two components are required for submission: a complete model for optimization and an observed streamflow data file. The complete model, obtainable via the aforementioned "Basin Extraction" and "Driving Data Preparation" modules, is a compressed archive containing all necessary input data and the executable. The observed streamflow data file should be a single-column text file, with a data series length that matches the model's simulation period. For missing data points, −999 should be used as a placeholder; these values will be excluded from the calculation of evaluation metrics during the optimization process.

Upon initiating the optimization program, the system automatically performs the calibration, logging the parameter combinations, corresponding NSE values, and optimization progress for each iteration to an output file. This information is relayed to the user through a service interface, enabling real-time monitoring of the calibration progress.

3.2.4. One-click modeling

To further enhance the efficiency of hydrological modeling, this platform integrates the three core steps—basin extraction, driving data generation, and model parameter optimization—into a single, automated workflow, providing users with a one-stop modeling experience. Users begin by completing a detailed parameter form on the front-end interface, which covers the requirements for all three modules. Upon submission, the platform initiates the automated process, with a progress bar displaying the real-time status of each module to allow users to track the workflow. Once basin extraction is complete, the results are instantly visualized on the map, which automatically triggers the driving data generation task. Following its completion, the system seamlessly transitions to the parameter optimization phase. During this phase, the front-end charts are periodically updated to display both the observed and simulated streamflow hydrographs, enabling users to intuitively observe the dynamic changes in the simulation performance and evaluation metrics, thereby gaining an in-depth understanding of the optimization process. Upon successful completion of all tasks, the platform generates a comprehensive modeling package for download. This archive contains all necessary resources, including the basin's spatial files, driving data, optimized parameters, and the model executable, ensuring that all outputs can be easily accessed and reused.

3.2.5. Backend results download

The design philosophy of this online modeling platform is to provide a rapid and convenient user experience, minimizing the operational burden and time investment for users. To avoid requiring users to remain online for extended periods, the platform employs an asynchronous task processing model, analogous to a modern delivery service. After submitting a task, users can safely leave the platform and retrieve their results later using a unique "download code." Upon task

Table 3
Available integrated datasets for model inputs.

Data type	Dataset name	Spatial resolution	Temporal resolution	Coverage area	Coverage period	Source
Precipitation	GPM_IMERG (Final Run)	0.1°	Daily	Globe	2000/06/01–2021/08/31	Huffman et al., 2019
	ERA5	0.25°	Daily	Globe	1980/01/01–2024/12/31	Hersbach et al., 2020
	ERA5_Land	0.1°	Daily	Globe	1950/01/01–2021/11/30	Muñoz-Sabater et al., 2021
	CMFD	0.1°	Daily	China	1979/01/01–2018/12/31	He et al., 2020
Evaporation	ERA5	0.25°	Daily	Globe	1980/01/01–2024/12/31	Hersbach et al., 2020
	ERA5_Land	0.1°	Daily	Globe	1950/01/01–2024/08/24	Muñoz-Sabater et al., 2021
Temperature	ERA5	0.25°	Daily	Globe	1980/01/01–2024/12/31	Hersbach et al., 2020
	ERA5_Land	0.1°	Daily	Globe	1950/01/01–2020/12/31	Muñoz-Sabater et al., 2021
NDVI	NDVI_MOD13A2	500 m	Monthly	Globe	2000/01/01–2019/12/31	Myneni et al., 2021
LAI	LAI_MOD15A2H	500 m	Monthly	Globe	2000/01/01–2019/12/31	Didan, 2021
Soil	Soil	30″	–	Globe	–	Dai et al., 2019

submission, users can not only monitor the progress in real-time via the front-end interface but also immediately receive a text file containing the download code and detailed task information.

To ensure uniqueness and security, the download code follows the format "X-yyyymmddHHMMSSabcd". The leading digit "X" (ranging from 1 to 4) indicates the specific task type (1: Basin Extraction; 2: Zonal Statistics; 3: Driving Data Generation; 4: Parameter Calibration). This is followed by a timestamp precise to the second of the task submission, and a four-digit random number, guaranteeing the uniqueness of each code.

All tasks are executed in separate directories named after their respective download codes, ensuring orderly and isolated data processing. The front-end interface displays the task progress in real-time, and closing the browser does not interrupt the ongoing execution of the task on the server. Users can query the task status at any time using the download code. If the task is completed, the platform provides a download link for the final product, typically a compressed archive. If the task is still in progress, the current progress is displayed.

3.2.6. Model sharing

To foster communication and collaboration among users, the platform features a dedicated "Model Sharing" module. This module enables users to share their hydrological models and provides a convenient channel for others to download and utilize these shared resources.

Through this module, users can browse a list of all publicly shared models. The list displays key information for each model, including its unique identifier, name of the associated basin, time range, and accuracy evaluation metric (NSE). With a single click, users can download a model and access its detailed information. Concurrently, the platform offers a user-friendly interface for model sharing, encouraging users to contribute their research outcomes. To share a model, a user must provide a name, specify the coverage period, and upload the model package (including the executable and input/output data) along with a geographic information file for visualizing its location on the platform map. Additionally, a brief description can be attached to clarify the model's background and applicability.

It is anticipated that this module will enhance the efficiency of data circulation and establish an open, interactive platform of high quality for the research, validation, and application of hydrological models.

3.2.7. LLM agent development and integration

The agent for this study was developed on the ChatGLM platform. This platform offers a user-friendly framework for agent creation, enabling developers to rapidly prototype agents and thereby dedicate more focus to critical tasks such as enhancing the knowledge base and optimizing prompts.

The primary step in developing a GLM agent is crafting the prompt (i.e., the configuration information). As the core instruction that drives Large Language Models (LLMs), the quality of the prompt directly determines the effectiveness of the model's output. The agent utilizes this prompt to parse user inputs, comprehend semantic intent, and generate corresponding responses. Therefore, a precise prompt description is crucial for ensuring the output meets expectations.

The primary purpose of the configuration information is to define the agent's role, objectives, and task workflow. To ensure clear and effective instructions, this study employed the BRTR optimization framework (GLM Team, 2026) for prompt engineering. This framework implements instruction tuning through a four-step progressive strategy: first, establishing a cognitive map of the problem domain in the **Background** layer; then, defining the agent's professional identity in the **Role** layer; subsequently, decomposing specific operational procedures in the **Task** layer; and finally, defining output standards in the **Request** layer. This hierarchical and progressive configuration method effectively addresses common issues in traditional prompt engineering, such as intent drift and uncontrolled output.

The detailed configuration information used in this study is provided in the Supplementary Materials. Critically, we emphasized the agent's three primary tasks: (1) To answer a wide range of user questions, providing professional responses, particularly on hydrological topics; (2) To provide user guidance for the HydroCraft Platform based on the knowledge base; (3) To comprehend user requirements for building hydrological models (including basin extraction, driving data preparation, and parameter calibration) and generate the request body required by the platform's server in a standardized format.

The second and most labor-intensive phase of agent development is the construction of the knowledge base. The knowledge base for this platform's agent comprises the following files (see Supplementary Materials for details):

Agent Work Standards.txt: This file serves as the agent's "code of conduct," acting as both its "job description" as a team member and a "service agreement" with the developers. It elaborates on the agent's primary duties and performance standards, reinforcing and specifying the configuration information.

User FAQ.txt: This file contains a collection of potential user questions and their corresponding answers. In this study, over 100 students from an "Advanced Hydrology" course were invited to build hydrological models using the HydroCraft Platform, and their feedback was collected. While some technical issues were resolved, other common user questions were compiled into this file, such as "What does the 'Catchment Area Threshold' on the basin extraction interface mean?", "How do I define the basin boundary?", and "What should I do if the model executable crashes upon launch?".

China Hydrological Stations Basic Database.xlsx: This file catalogs detailed information for over a thousand major hydrological stations in China, including station ID, name, basin, and geographic coordinates. This not only provides users with comprehensive reference data but, more importantly, allows the agent to quickly query and confirm basin outlet coordinates when a user provides just a station or basin name. This supports users in completing operations like basin extraction and lays an informational foundation for constructing hydrological models via natural language.

HydroCraft Platform User Manual. pptx: This document contains detailed modeling steps, designed to guide users through the hydrological model construction process on the platform.

Models Descriptions.docx: This document provides detailed descriptions of the THREW model, the Miyun Hydrological Model, and the Xin'anjiang Model, covering their principles, input/output file formats, and parameter adjustment methods.

The knowledge base will be continuously updated and enhanced with additional relevant files.

After uploading the knowledge base files, the agent undergoes debugging to test its performance. The performance of the ChatGLM agent is primarily influenced by three key elements: the configuration instructions (prompt), the content of the knowledge base, and the "generation diversity" parameter. The "generation diversity" parameter (range 0–1) controls the variability of responses: a value closer to 0 yields more stable and deterministic answers, while a value closer to 1 allows for more diverse and creative outputs. The agent's responses are most accurate and helpful when the configuration instructions and the knowledge base content are precisely aligned with potential user queries and their correct answers. The debugging process, therefore, involved iteratively simulating various user query scenarios, and continuously refining the configuration instructions, expanding/refining the knowledge base content, and adjusting the "generation diversity" parameter. This iterative cycle aimed to eliminate contradictions, achieve an appropriate balance between conciseness and detail (or between stability and creativity) in responses, and ultimately ensure that the agent's replies reliably and accurately meet user needs. Once ready, the agent is published and integrated into the HydroCraft platform via the ChatGLM WebAPI.

The LLM agent developed through the aforementioned process can provide extensive knowledge in the field of hydrology, guide users in

building models on this platform, and answer questions they encounter. However, at that preliminary stage, the agent functioned primarily as a knowledgeable chatbot and lacked the ability to execute operational tasks autonomously. To further enhance the platform's modeling efficiency and level of intelligence, a strategy of deep integration between the agent and the modeling platform was adopted. This integration enables the agent to discern user intent and proactively execute corresponding operational tasks on the platform.

As previously mentioned, the configuration information specifies one of the agent's tasks: to understand user requirements for building a hydrological model (e.g., basin extraction, driving data preparation, parameter calibration) and generate a request body in the standard format required by the platform's server. This task is further detailed and standardized in the knowledge base file "Agent Work Standards." Once configured, when a user makes a specific request, the agent will generate a response containing a special identifier (a JSON object). The platform's front-end monitors these responses in real-time; upon detecting one with this identifier, it parses the JSON content and sends a request to the server to execute the corresponding task (as shown in the Fig. 6).

This identifier has two key characteristics. First, the agent only generates a response containing a JSON request body in specific scenarios, which distinguishes it from conventional interactions. Second, the standardized JSON content includes two attributes: "purpose" and "content." The "purpose" attribute indicates the specific user intent parsed by the LLM agent. When the value of "purpose" is a valid command recognized by the platform, the front-end controller invokes the corresponding method based on this value, using the "content" attribute as the parameter to send a request to the server, while simultaneously displaying real-time information such as progress on the user interface.

For instance, in a scenario where a user provides a hydrological station name or coordinates to perform a basin extraction, the specification for the agent's response is as follows:

Upon recognizing a user's intent to perform basin extraction, the agent should actively inform the user of the three available methods. …… Once the longitude and latitude coordinates of the hydrological station are known, the agent must generate a JSON text in the format below and present it to the user in standard JSON format. Note that only the values within the {} should be replaced with the user-provided (or looked-up) longitude and latitude coordinates; all other content must remain unchanged.

```
{
  "purpose": "basin_extract",
  "content": {
    "dem_Dataset_Name": "MeritHydro",
    "box_extent": "",
    "outletXY": "{longitude},{latitude}",
    "snapDistance": "0.05",
    "minAccArea": "0",
    "minSubbasinArea": "0",
    "maxSubbasinArea": "0",
    "clipFeatureFile_zip": ""
  }
}
```

Thus, when a user requests basin extraction, the agent generates a response containing a JSON request body as a specific identifier. The front-end detects this identifier, parses the JSON, and sends the corresponding request to the back-end to execute the basin extraction task.

3.2.8. Visualization

This module is built upon the VUE3 frontend framework, integrating the OpenLayers mapping engine and ECharts charting library to provide comprehensive visualization capabilities, ranging from spatial data rendering to time-series analysis.

As a modern frontend framework, VUE3 is renowned for its efficient reactive system and Composition API, which significantly enhances development efficiency and code maintainability. Within the platform, VUE3 manages the application's state and routing and facilitates modular development through a component-based architecture. For instance, core functionalities such as maps, charts, and forms are encapsulated as independent components, promoting reusability and extensibility.

The user interface components are built with Element Plus, a Vue 3-based UI component library that offers a rich set of pre-built components (e.g., forms, buttons, dialog boxes) and a modern design aesthetic.

The map visualization functionality relies on the synergy between GeoServer and the OpenLayers mapping engine. GeoServer, an open-source geospatial data server, supports publishing map data via the WMS (Web Map Service) and WFS (Web Feature Service) protocols. In this platform, spatial data such as river networks, hydrological stations, and basin boundaries are published as standardized services through GeoServer, ensuring efficient data transmission and dynamic updates. OpenLayers serves as the front-end mapping engine, responsible for

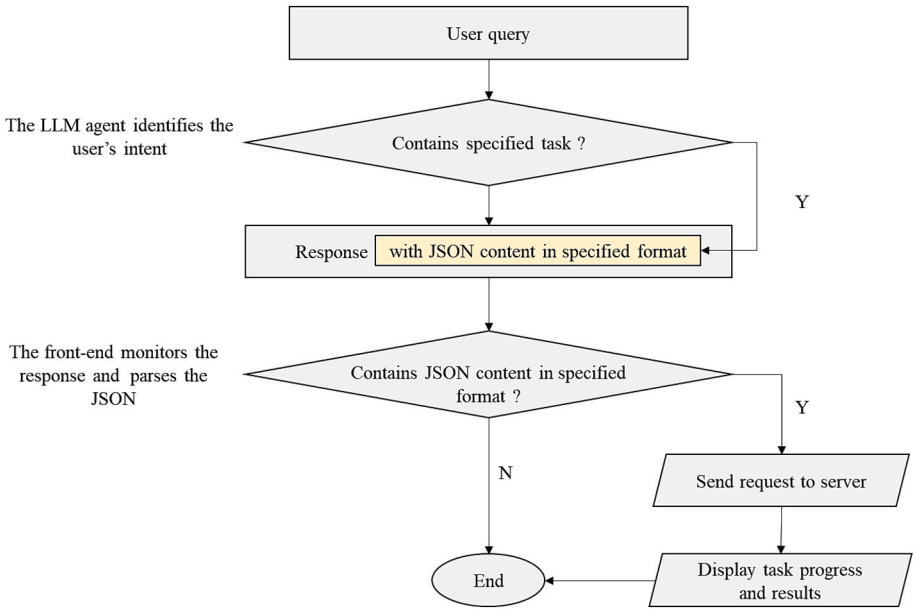


Fig. 6. Schematic of the integration mechanism between the modeling platform and the LLM Agent.

loading and rendering these map services from GeoServer, as well as spatial data generated by the back-end. It also provides a rich set of interactive features, including zooming, panning, and querying.

4. Case study

4.1. Constructing a hydrological model using the platform interface

This section demonstrates how to quickly build a hydrological model using the Jiahe Hydrological Station, located on the Chunling River (a tributary of the Xiang River in China), as an example.

Click the "One-Click Modeling" icon to enter the modeling interface. The specific operational interface and workflow are illustrated in Fig. 7.

Step 1: Basin extraction.

For the basic setup, the user only needs to provide the coordinates of the Jiahe Hydrological Station (112.3667, 25.5667). The coordinates can be entered directly or by locating the station on the map and clicking, which will auto-populate the input field.

For a more customized model, additional settings can be configured:

DEM Data Source: Select MERIT_DEM_90m.

Basin Boundary: Use the map box selection tool (when drawing the rectangle, refer to the river network layer and make it sufficiently large to enclose the entire basin of interest), with the input value "112.0297,112.5240,25.6181,25.0717".

Outlet Snap Distance: Set to 0.01.

Threshold Drainage Area: Set to 50.

Min/Max Sub-basin Area: Set to 20 and 150, respectively.

Step 2: Driving data generation.

Set the date range from January 1, 2015, to December 31, 2015. Select GPM_IMERG_Final for precipitation data, and leave other options as default.

Step 3: Upload measured flow data file to calibrate the hydrological model.

After completing the above steps, click the "Submit" button. The front-end will sequentially display the progress and results of each modeling stage (as shown in Fig. 8). First, the basin extraction task is executed, with its progress shown in Fig. 8(a). Upon completion, the extracted basin is displayed on the map, and the driving data preparation task begins, with its progress shown in Fig. 8(b). After this stage, the parameter optimization task is initiated, with the front-end displaying

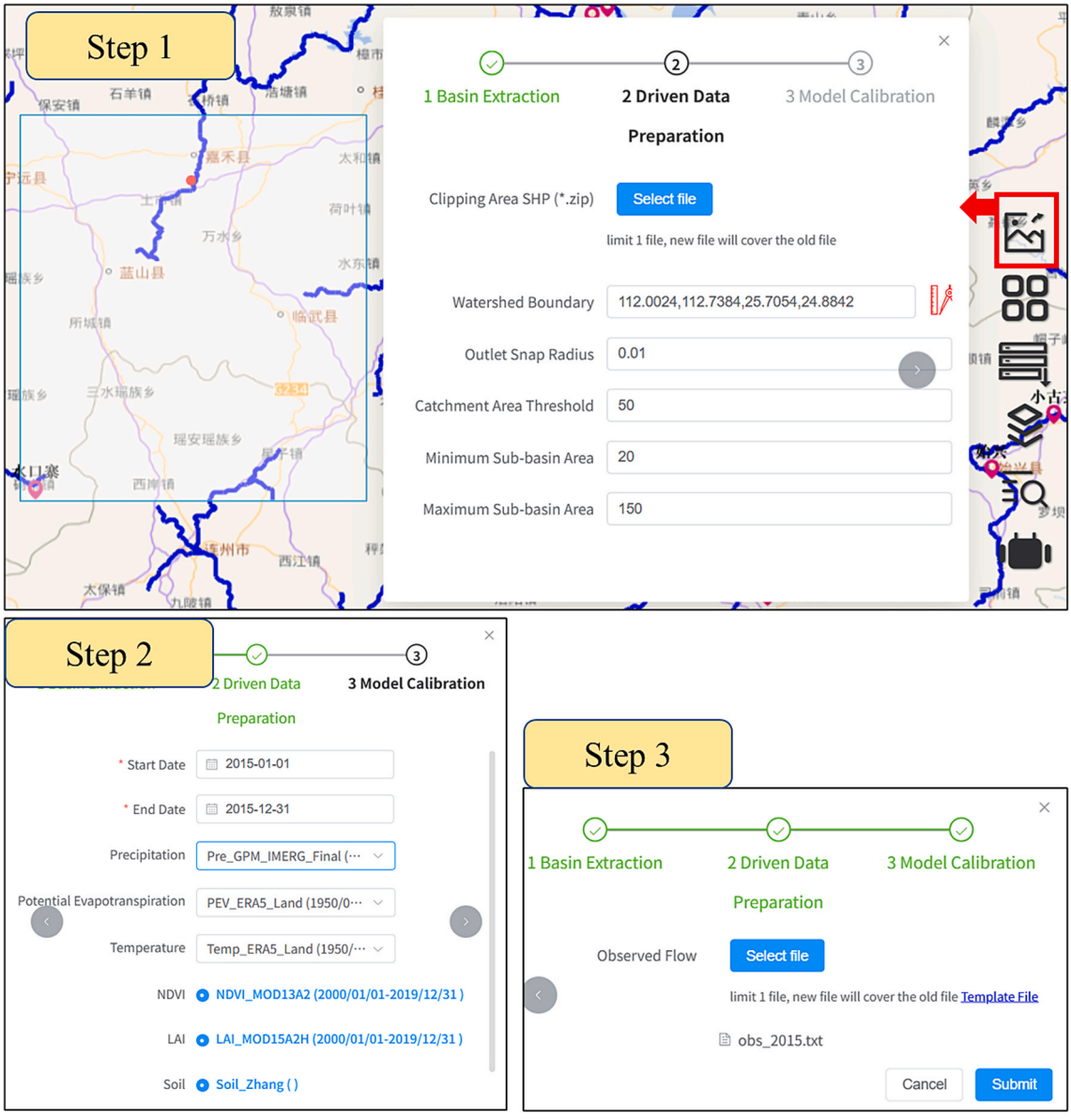


Fig. 7. Operational flow of the one-click modeling feature in the HydroCraft platform.

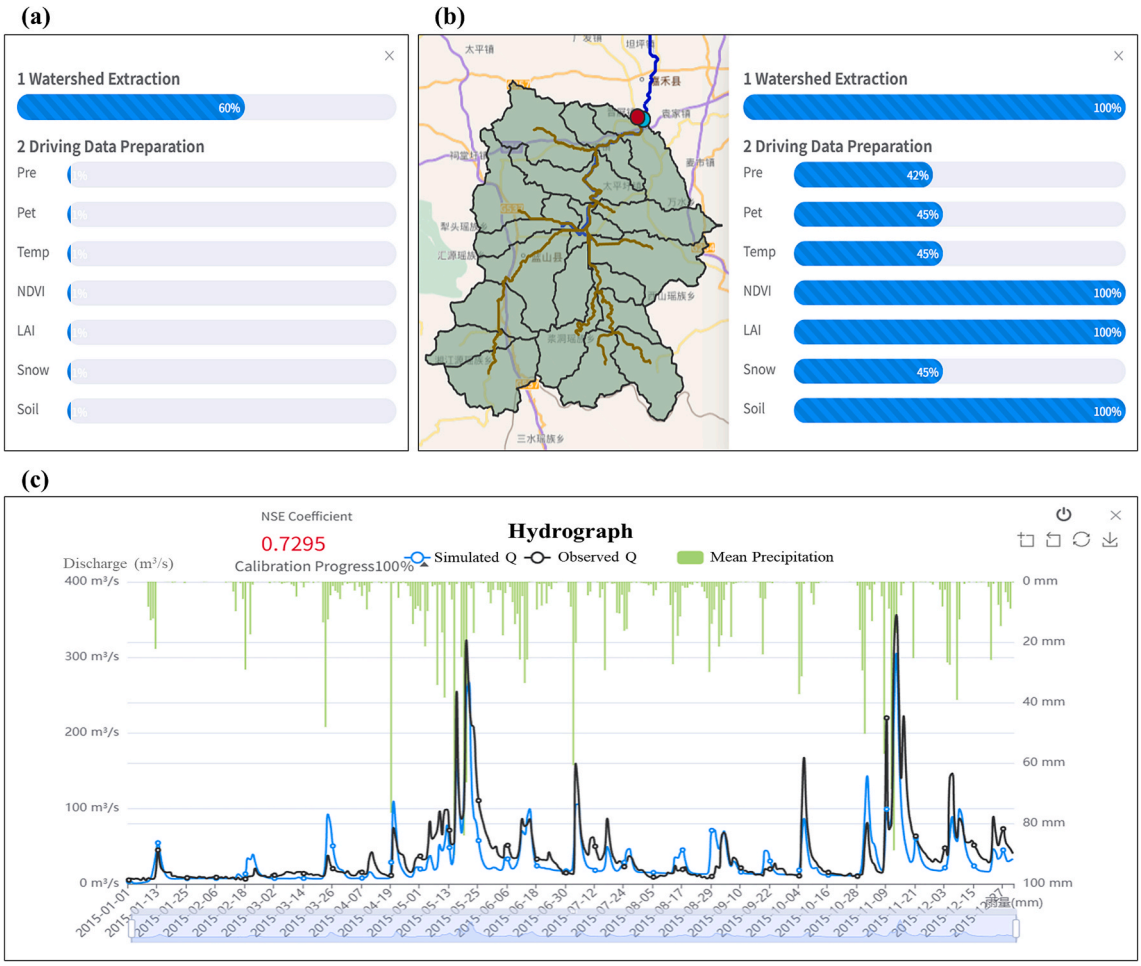


Fig. 8. Progress and results at each modeling stage on the front-end interface.

the optimization process in near real-time, including the observed and simulated streamflow hydrographs, the NSE metric, and the progress percentage. In this case, the final NSE value was 0.7295, and the total modeling process took approximately 6 min.

4.2. *Constructing a hydrological model using the LLM agent*

This section continues to use the upstream basin of the Jiahe Hydrological Station as a case study to demonstrate the construction of a hydrological model through natural language interaction with the LLM agent, and to illustrate the other significant roles the agent plays throughout the modeling process.

By initiating the agent's dialogue box, the user can issue a command, either by voice or text, such as, "Please extract the upstream basin of the Jiahe Hydrological Station." The agent responds with a JSON payload, which automatically triggers the platform's "Basin Extraction" task. Upon completion, a compressed file of the extraction results is delivered to the user's client. The successful basin extraction lays the foundation for the model; the subsequent steps involve replacing the driving data and performing parameter optimization as needed. While these two steps involve complex operations like file uploading, which are currently beyond the agent's direct capabilities, the user can continue to consult the agent for operational guidance on data preparation and parameter optimization. Partial interfaces of these operations are shown in Fig. 9. It is worth noting that the agent supports multiple mainstream languages, including Chinese, English, French and Arabic to accommodate a global user base.

Agent-assisted modeling is a novel approach that significantly lowers the barrier to entry for users. Notably, another benefit of incorporating the agent is its potential to enhance model performance. The agent leverages a knowledge base, which accumulates the modeling experience and expertise of developers and other users, to provide valuable suggestions for improving model accuracy. For instance, in the case of the Jiahe upstream basin, although the hydrological model performed well overall, it underestimated the peak flow, resulting in a significant relative error. To address this issue, the user consulted the agent for improvement suggestions (as shown in Fig. 10). The agent not only guided the user on how to adjust parameters to enhance the simulation accuracy for the peak flow but also warned of the potential risk of degraded overall performance from localized parameter tuning. Following the agent's recommendations, moderate adjustments were made to the Muskingum parameters. As a result, the model's performance in simulating the peak flow was effectively improved, and the relative error of the flood peak was reduced.

5. Limitations and future developments

To enhance modeling efficiency, we integrated the critical steps required for hydrological modeling and incorporated an LLM agent to assist users in task execution and query resolution. Despite these efforts, the current platform still has several limitations. First, the variety of tasks that the agent can autonomously execute remains limited. Its task-execution capability depends on a carefully curated knowledge base and a small set of structured commands recognized by the front end. Our testing showed that simply expanding the knowledge base is not always beneficial: excessive detailed material can make the agent overlook

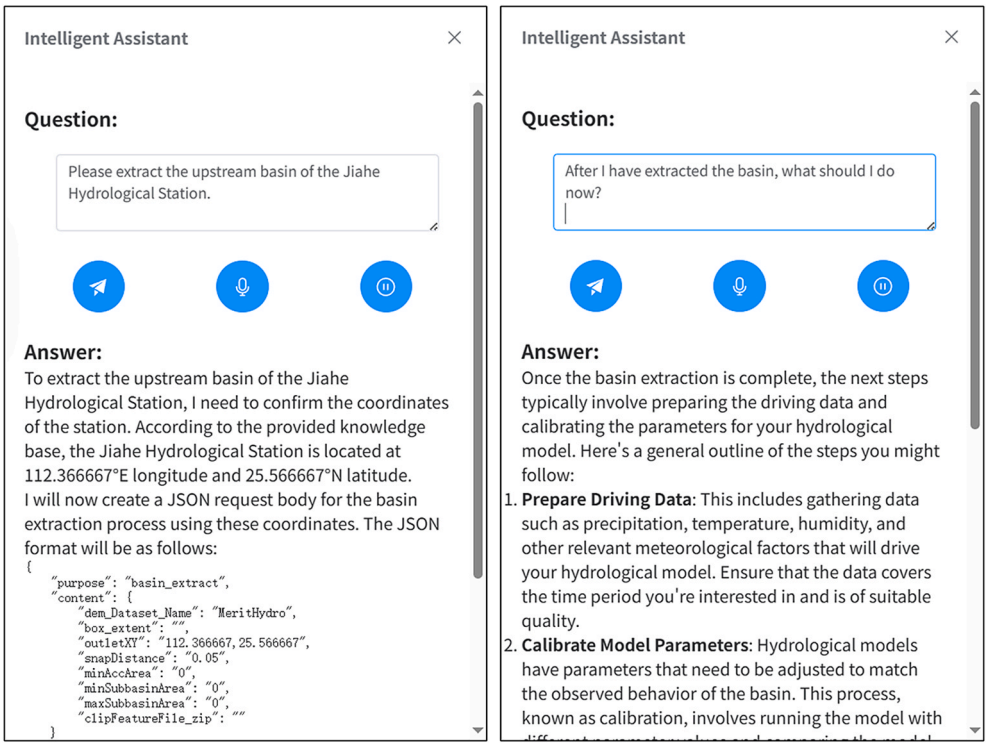


Fig. 9. Key interface snapshots illustrating the agent-assisted hydrological modeling workflow.

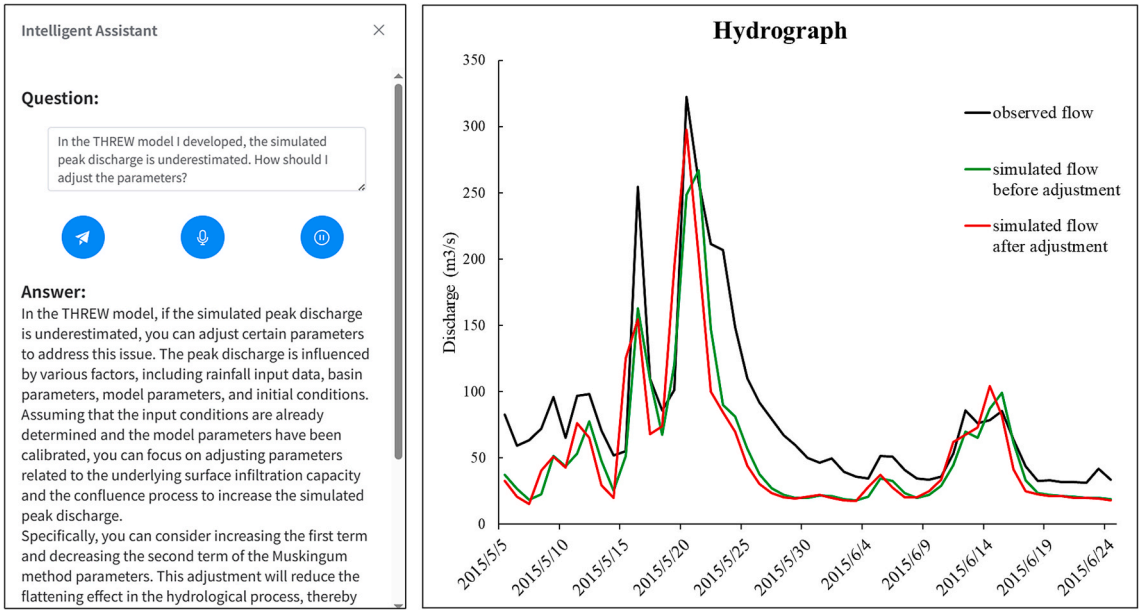


Fig. 10. Agent's parameter adjustment recommendations and the resulting model performance.

concise but critical instructions, whereas an overly small knowledge base reduces coverage. Consequently, the current knowledge base contains coordinate information mainly for hydrological stations in mainland China, so name-based modeling is currently supported primarily for those stations. Users can still apply the agent globally by providing explicit outlet coordinates, but broader name-based support requires additional data curation, retrieval rules, and validation.

Second, the occurrence of hallucinations is a well-known challenge for general-purpose LLMs. The foundational model used in this work has not been fine-tuned specifically for the hydrology domain, and the application of LLMs in hydrology and Earth science remains relatively new, with its limitations requiring systematic exploration. Even when RAG is used, LLM agents may produce plausible but incorrect explanations, misinterpret hydrological terminology, confuse coordinates or station names, generate outdated advice, or return structured requests that are syntactically valid but physically inappropriate. These risks are particularly important in hydrological modeling because small errors in basin outlets, forcing data selection, parameter ranges, or calibration periods can propagate into large errors in simulated streamflow. HydroCraft reduces these risks by grounding answers in a private knowledge base, restricting executable actions to standardized JSON requests, and keeping users in the loop; however, the agent should still

be treated as an assistant rather than an autonomous validator of hydrological correctness.

Furthermore, while the integrated hydrological models have been applied in many watersheds and tested by us in several specific ones (see Appendix B), their applicability across global watersheds still requires further validation. A key limitation common to the current HydroCraft models must be emphasized: models using default parameters or parameters suggested by the agent are not suitable for reliable application without calibration and evaluation. All three integrated models contain conceptual parameters whose values should be calibrated against observed streamflow data whenever such data are available.

Future work will therefore focus on both platform capability and scientific reliability. We aim to enrich and better structure the agent knowledge base, optimize agent integration through methods such as the Model Context Protocol (MCP), expand the range of executable tasks, and add automatic checks for units, outlet locations, data availability, parameter bounds, and physically unrealistic outputs. We will also integrate other representative and adaptable hydrological models. For large-scale basins with many sub-basins, we will further improve computational efficiency, uncertainty communication, and reproducibility. Finally, the model-sharing module will be used to support parameter regionalization, transfer learning, and community benchmarking so that HydroCraft can better serve both gauged and ungauged applications.

6. Summary and conclusions

This paper has detailed the development and implementation of HydroCraft, an intelligent platform that effectively addresses key limitations in contemporary hydrological modeling tools. The main conclusions and innovations are summarized as follows:

1) Full Workflow Automation and Integration: HydroCraft integrates the entire hydrological modeling chain—watershed delineation, data preparation, and parameter calibration—into a unified, web-based environment. All inputs to the embedded hydrological models, such as topography, meteorology and soil, are automatically created from publicly available databases and converted into input formats readable by the model. The platform automates the hydrological modeling process, drastically reducing the time and expertise required, as evidenced by the case studies where a model was built and calibrated in about six minutes.
2) A Novel Modeling Paradigm with LLM Integration: A cornerstone innovation of HydroCraft is its deep incorporation of a LLM agent, which establishes a new paradigm for interactive hydrological modeling. This approach moves beyond conventional graphical user interfaces (GUIs) and simple chatbots, shifting the interaction paradigm to a natural language-driven, agent-mediated workflow. The agent's capability to comprehend user intent and autonomously orchestrate the modeling process—from parsing commands like "extract the upstream basin" to generating standardized JSON requests that seamlessly trigger backend functions—represents a shift in human–computer interaction for scientific modeling. This agent-centric workflow significantly lowers the technical barrier, allowing users to focus on their scientific goals rather than operational complexities.
3) Access to Expert Knowledge and Enhanced Performance: The LLM agent, powered by a curated knowledge base, acts as an always-available expert. It not only guides users through operational steps but also provides actionable advice for model improvement. The case study confirmed that the agent's parameter recommendations could effectively enhance simulation accuracy, such as by improving peak flow predictions.
4) Promotion of Collaboration through Model Sharing: The platform includes a dedicated module for sharing hydrological models, fostering a collaborative community where users can disseminate and validate research outcomes, thereby accelerating the advancement of hydrologic science.

In summary, HydroCraft demonstrates a viable path toward more intelligent, efficient, and democratized hydrological modeling. By synthesizing automation, advanced AI, and user-centric design, it overcomes the traditional barriers of complexity and inaccessibility. Future work will focus on expanding the platform's knowledge base, incorporating additional hydrological models, and enhancing the multilingual capabilities of the LLM agent to further solidify its global applicability and impact.

CRediT authorship contribution statement

Mahmut Tudaji: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Conceptualization. **Fuqiang Tian:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Xiannan Wu:** Writing – review & editing, Validation, Resources, Methodology.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Fuqiang TIAN reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary evaluation and sensitivity analysis of the adaptive watershed extraction method

A.1 Study catchments and experimental design

To supplement the evaluation of the adaptive watershed extraction method and examine the sensitivity of the user-defined minimum and maximum sub-basin area parameters, five representative catchments were selected for supplementary experiments. These include the mountainous Xiahui catchment, the Zhangjiafen catchment with internal lakes, the plain catchment Huashan, the plateau catchment Tangnaihai, and the Montcalm basin in Michigan, USA, which flows to a lake. Basic information for the five catchments is summarized in Table A1, and their geographic distribution is shown in Fig. A1.

Table A1
Basic information of the five representative study catchments.

Basin feature	Outlet location name	Abbreviation	Outlet coordinates
Mountainous	Xiahui	XH	117.17, 40.62
With internal lakes	Zhangjiafen	ZJF	116.78, 40.61
Plain	Huashan	HS	120.83, 29.70
Plateau	Tangnaihai	TNH	100.15, 35.50
Flows to a lake	Montcalm	–	−86.23, 43.07

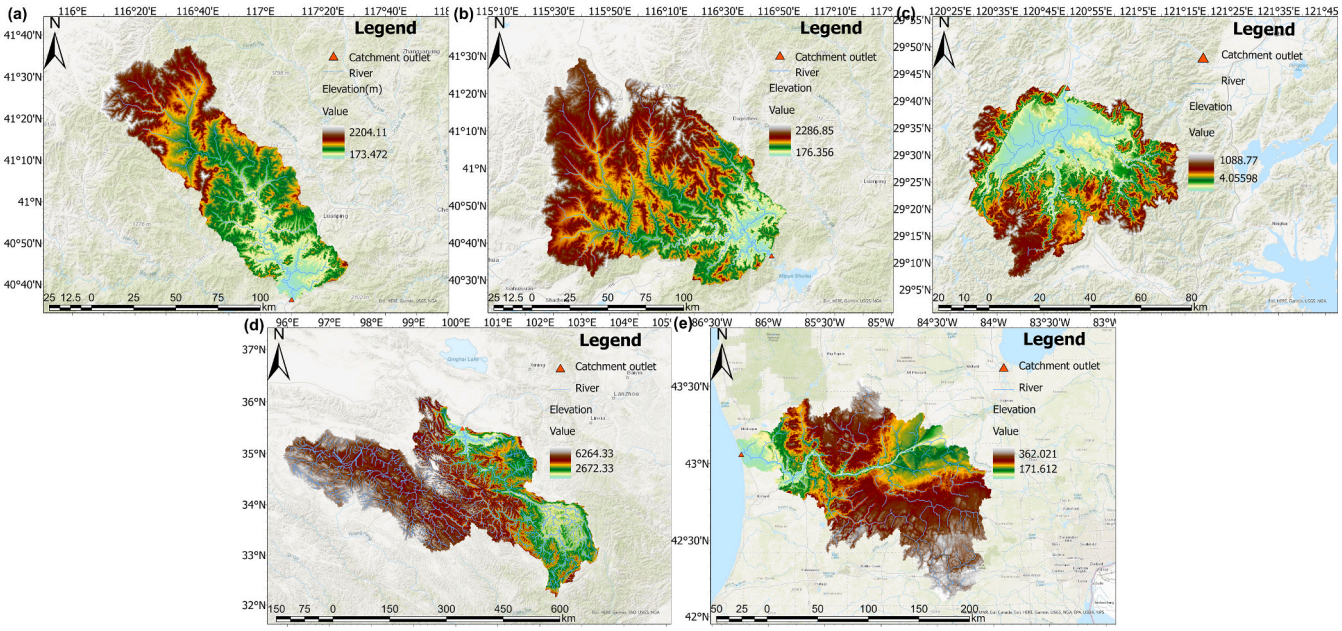


Fig. A1. Geographic distribution of the study catchments.

The main purpose of the adaptive watershed extraction method is to adaptively adjust the initial delineation result using user-defined lower and upper bounds for sub-basin area, thereby avoiding excessively large or excessively small computational units and promoting a more balanced discretization structure. By contrast, the conventional method only relies on the catchment area threshold to derive the river network and sub-basins, without imposing any explicit constraints on sub-basin area. Based on this methodological distinction, three paired comparison groups were designed under three catchment area threshold levels, and in each group the adaptive watershed extraction method was directly compared with the corresponding unconstrained conventional method.

Specifically, in the first experiment group, the flow accumulation area threshold (minAccArea) was set to 50 km^2, while the minimum and maximum sub-basin areas for the adaptive watershed extraction method were set to 10 km^2 and 100 km^2, denoted as scheme "50_10_100"; the paired conventional scheme without min/max constraints was "no_minmax_50". Similarly, the second and third groups employed thresholds of 100 km^2 and 200 km^2 with bounds of 20–200 km^2 and 40–400 km^2, corresponding to schemes "100_20_200"/"no_minmax_100" and "200_40_400"/"no_minmax_200", respectively. Summary statistics of sub-basin areas for all schemes are presented in Table A2, and the corresponding violin plots are shown in Fig. A2.

A.2 Evaluation criteria

Since the direct objective of the adaptive watershed extraction method is not to alter the total watershed area but rather to regulate the scale distribution of the delineated sub-basins, the present evaluation focuses on sub-basin area distribution characteristics. Three summary statistics were used: minimum sub-basin area, maximum sub-basin area, and mean sub-basin area. In addition, violin plots were used to visually compare the concentration, dispersion, and overall range of sub-basin area distributions under different schemes. This evaluation framework directly examines two questions: first, whether the adaptive watershed extraction method can effectively satisfy the prescribed lower and upper area bounds; and second, whether the adaptive watershed extraction method yields more uniform and controllable computational units than the conventional method across different catchment types and threshold settings.

A.3 Results and discussion

Table A2
Summary statistics of sub-basin areas for the five catchments under six watershed extraction schemes.

Schemes	Value	XH	ZJF	HS	TNH	Montcalm	Average
50_10_100	min	11.12	10.57	11.22	10.05	10.20	10.63
	max	99.14	98.90	97.90	99.95	99.74	99.13
	average	58.88	56.34	56.80	56.67	56.00	56.94
100_20_200	min	22.25	22.45	21.45	20.36	23.26	21.96
	max	194.72	195.86	191.44	199.70	193.85	195.11
	average	102.33	111.26	113.60	111.96	107.80	109.39
200_40_400	min	49.87	49.08	97.65	40.88	45.26	56.55
	max	359.53	387.51	364.83	398.48	396.10	381.29
	average	193.11	209.27	255.61	236.29	228.19	224.49
no_minmax_50	min	1.18	0.04	1.33	0.01	0.19	0.55
	max	266.16	294.73	221.10	665.44	264.10	342.30
	average	72.06	77.78	71.33	69.87	67.49	71.71
no_minmax_100	min	2.78	0.06	1.33	0.02	1.48	1.13
	max	283.95	413.22	533.90	836.08	479.75	509.38
	average	117.75	139.51	133.36	143.62	122.87	131.42
no_minmax_200	min	2.78	1.07	1.33	0.32	7.96	2.69
	max	404.47	1015.12	885.19	1884.27	1543.14	1146.44
	average	178.81	237.55	340.81	319.20	281.89	271.65

The results show that under all three parameter groups, the adaptive watershed extraction method consistently constrained the delineated sub-basin areas of the five catchments within the prescribed ranges, demonstrating robust parameter responsiveness and effective enforcement of the user-defined bounds. In contrast, the conventional method exhibited substantially greater dispersion and a much larger area range in all three comparison groups. A representative example is the TNH catchment under "no_minmax_50", where the minimum sub-basin area was only 0.01 km^2 while the maximum reached 665.44 km^2. This indicates that the conventional method tends to generate both extremely small and extremely large units simultaneously, resulting in highly imbalanced sub-basin discretization.

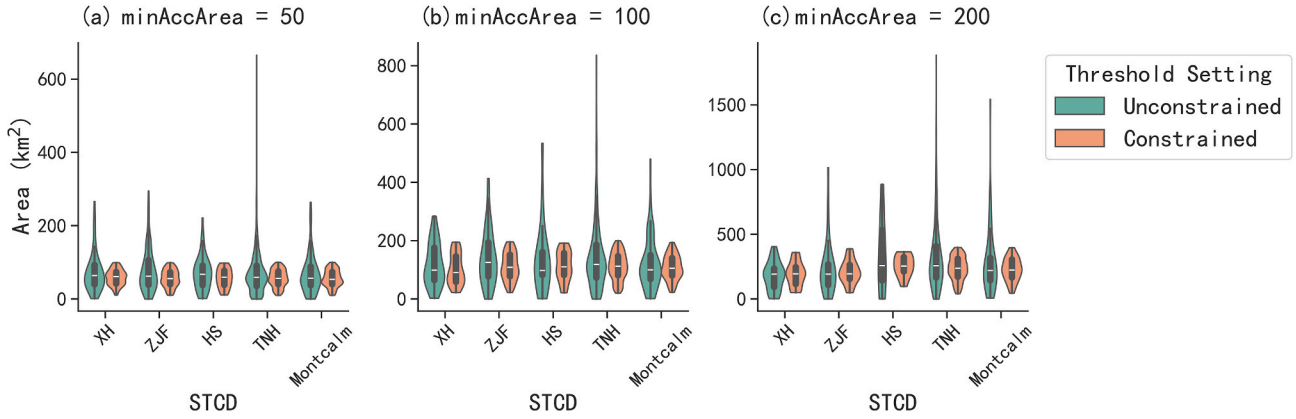


Fig. A2. Violin plots of sub-basin area distributions under the adaptive watershed extraction method and the conventional method for the three paired comparison groups.

Fig. A2 further reveal a clear and consistent pattern across all three experiment groups and all five catchments: the results derived from the adaptive watershed extraction method show more concentrated distributions, narrower ranges, and lower dispersion, whereas the conventional method yields more scattered distributions with longer tails, indicating a higher tendency to generate anomalously small or large sub-basin units. As the catchment area threshold and the corresponding minimum/maximum sub-basin area bounds increased simultaneously, the distributions derived from the adaptive watershed extraction method shifted toward larger scales in a predictable manner while remaining well bounded and stable. This indicates that the minimum and maximum sub-basin area parameters in the adaptive watershed extraction method are not arbitrary add-ons; rather, they function as effective control variables that regulate the sub-basin scale structure in a stable way. Although the present supplementary experiment did not directly test the influence of different delineation results on runoff simulation accuracy, a more uniform and controllable discretization structure provides a more reasonable basis for subsequent hydrological simulation, parameter regionalization, and computational efficiency improvement.

Appendix B. Global applicability test and model performance evaluation

To assess the global applicability of the modeling platform and evaluate its performance under both ungauged and gauged (with observed streamflow) conditions, the following tests were conducted.

Four watersheds located in different international regions and of varying scales were selected (their locations and basic characteristics are shown in Fig. B1 and Table B1, respectively). Hydrological models were constructed for these watersheds using the HydroCraft platform. The THREW model was selected as the model type. For the driving data, ERA5-Land data was chosen for precipitation, evaporation, and temperature, while other data options used the platform defaults.

Table B1
Basic information of the case study catchments.

Cases	River	Station	Country	Area (km^2)
case_1	Mekong	Stung Treng	China, Myanmar, The Lao, Thailand, Cambodia, Vietnam	640,876
case_2	Guanjiang	Guang Yang	China	952
case_3	Drava	Ortilos	Austria, Croatia, Hungary	30,789
case_4	Maroni	Maripasoula	Suriname, French Guiana	14,696

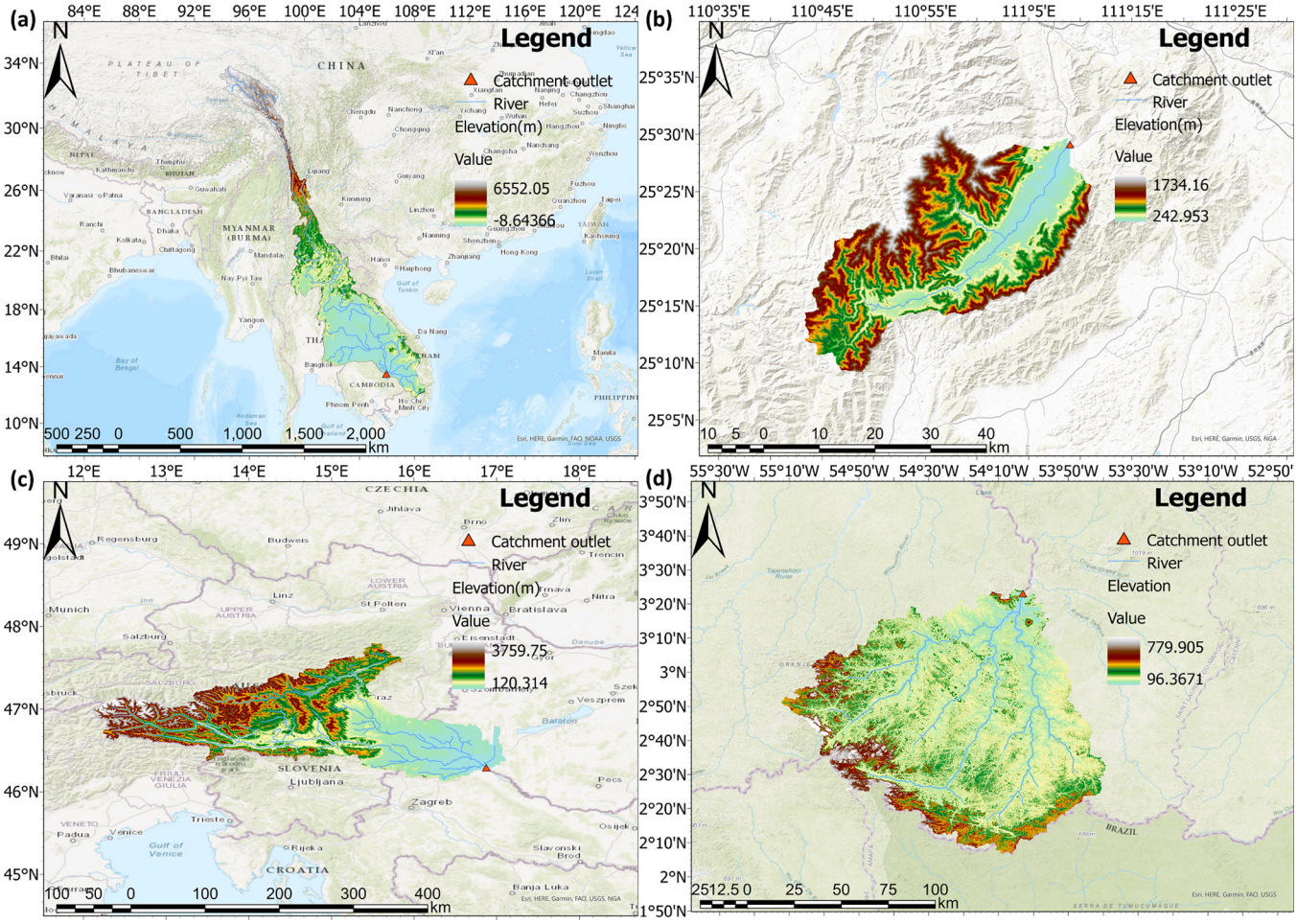


Fig. B1. Geographic distribution of the study catchments

The hydrological simulation results before and after calibration are presented in Fig. B2. The results indicate that during the calibration period, the model performance was reliable across all these watersheds, with particularly better simulation performance in larger watersheds characterized by lower flow variability. However, the model performance was poor under the uncalibrated condition (using the current default parameters of HydroCraft).

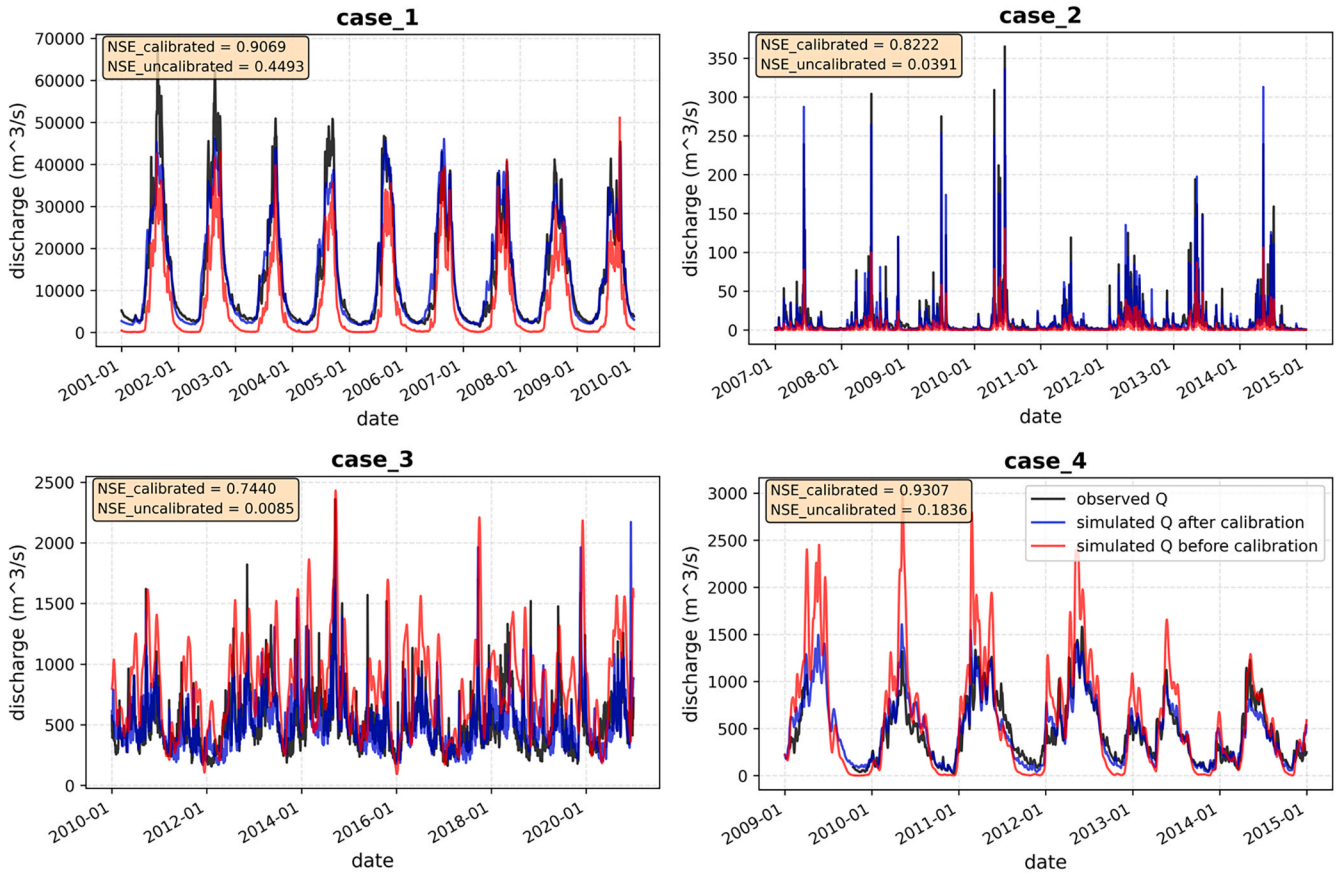


Fig. B2. Hydrological simulation results before and after calibration in the case catchments

To further evaluate the uncertainty under uncalibrated conditions, we extended a previous study (Tudaji et al., 2025b) by constructing THREW models for 63 small and medium-sized watersheds in southern China. The cumulative distribution functions of the Nash-Sutcliffe Efficiency (NSE) metric for simulations before and after model calibration are shown in Fig. B3. The results show that the models exhibited good performance after calibration (mean NSE of 0.84), indicating good applicability. However, the pre-calibration performance was generally weaker than the post-calibration performance, with a significant performance gap. Specifically, 44% of the watersheds had negative NSE values before calibration, highlighting the considerable uncertainty under uncalibrated conditions and the necessity of parameter calibration.

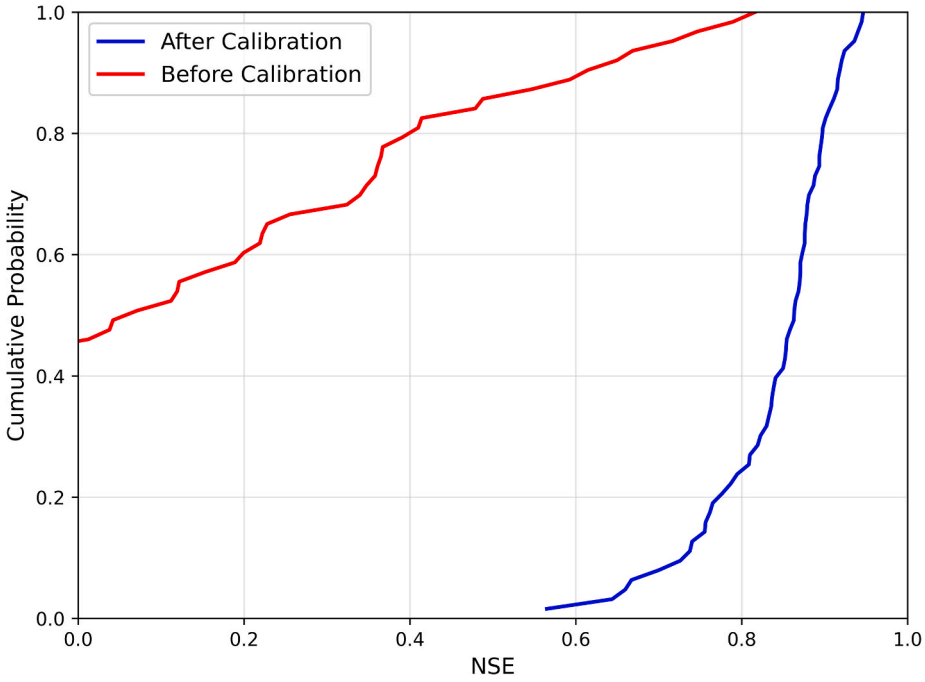


Fig. B3. The CDF plot of the NSE for simulations before and after model calibration.

The three models integrated into this platform have been applied in other studies (details in Section 2.2). This modeling platform provides a convenient service to facilitate their application to other watersheds. We have also tested them in several watersheds across different regions, but model performance remains influenced by various factors, including the quality of the driving data used, the selected spatial computational scale, and the data quality during the calibration period. Therefore, although we have attempted testing in a limited number of watersheds across different global regions and spatial scales, a truly comprehensive global-scale performance evaluation requires further research, which is part of our future plans. Nevertheless, a clear point is that all models currently integrated into the platform contain parameters that require calibration. The evaluation above also demonstrates the significant uncertainty in simulation results under uncalibrated conditions. Consequently, we strongly advise users to perform model calibration before formally applying the models for research or operational purposes.

Simultaneously, to overcome the challenge of application in data-scarce conditions, we encourage users to share their own models and data via the platform's model-sharing module. We envision that through collaboration and methods like parameter regionalization, we can collectively advance hydrological prediction in ungauged basins and contribute to solving water-related problems.

Data availability

Data will be made available on request.

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